INTELLIGENT DETECTION AND CLASSIFICATION OF AERIAL OBJECTS USING DEEP LEARNING

A Final Year Project Submitted by John Smith

Department of Computer Science • 2025

Abstract

This project presents an extensive investigation into the development, training, evaluation, and deployment

of a deep-learning-based aerial object classification system capable of distinguishing between drones, birds,

planes, and miscellaneous airborne entities. As global airspace becomes increasingly saturated with

commercial, private, and autonomous aerial vehicles, ensuring safety and situational awareness has never

been more critical. Conventional detection systems such as radar and acoustic sensors continue to struggle

with low-RCS (radar cross-section) targets such as consumer drones, which fly at low altitudes and often

share similar signatures with birds. Consequently, computer-vision-based detection has emerged as a

promising alternative approach.

This dissertation expands existing research by developing a fully reproducible framework that incorporates

modern CNN and transformer-based architectures, rigorously evaluated across multiple dimensions

including classification accuracy, inference latency, generalisation, dataset imbalance, robustness to

noise/environmental distortion, and interpretability through saliency mapping. The project uniquely

compares baseline custom CNN architectures with state-of-the-art pretrained models such as ResNet-50,

MobileNetV3-Large, and ConvNeXt-Tiny, evaluating both frozen and fine-tuned variants under identical

conditions. Additional focus is placed on the engineering challenges associated with dataset preparation,

including the construction of a unified aerial-object dataset from multiple sources, implementation of

preprocessing pipelines, augmentation strategies, and automated hyperparameter tuning.

The resulting system demonstrates that fine-tuned ConvNeXt-Tiny delivers exceptional accuracy, achieving

over 96% performance on the test set, while MobileNetV3-Large demonstrates outstanding inference speed

suitable for embedded real-time deployment. Extensive interpretability analysis reveals valuable insights

into model behaviour, highlighting the importance of shape, wing patterns, and rotor structures during

classification. Future expansion of the project may include YOLO-based detection, temporal tracking,

sensor fusion with radar, and deployment via NVIDIA Jetson or edge TPU hardware.

Chapter 1 – Introduction

The rapid rise in consumer and industrial drone usage has led to increasing concerns regarding airspace security, privacy, and public safety. Drones have become widely accessible, inexpensive, and capable of sophisticated autonomous operations. As airports, power stations, correctional facilities, and government bodies encounter growing incidents of drone intrusion, research into advanced classification systems has accelerated dramatically.

The primary purpose of this project is to construct a system capable of accurately classifying aerial objects from still images using state■of■the■art deep neural networks. The research involves not only constructing models but also understanding dataset limitations, environmental variability, and real■world constraints in deployment scenarios.

1.1 Motivation

The need for reliable drone detection is set against a global rise in unauthorised drone activity. Incidents include airport shutdowns, illegal border crossings, contraband delivery to prisons, and surveillance of sensitive locations. Traditional radar systems cannot reliably differentiate drones from birds due to similar radar signatures, creating a demand for optical classification solutions.

1.2 Research Questions

- How effectively can modern pretrained neural networks classify aerial objects compared to a custom-built CNN?
- What preprocessing and augmentation techniques best mitigate dataset irregularities?
- How do various optimisation strategies impact classification accuracy and throughput?
- How can interpretability methods reveal insights regarding feature importance?

1.3 Contributions

This dissertation contributes:

• a comprehensive dataset combining multiple public repositories,

• a full comparison across 7 model configurations,

• systematic evaluation against real-world noise,

• deployment-oriented inference benchmarks,

• an extensive interpretability study using Grad■CAM.

Chapter 2 – Literature Review

2.1 Overview

Deep learning has revolutionised image classification tasks, replacing classical techniques that relied on

hand■crafted features. CNNs automatically extract feature hierarchies, making them ideal for aerial object

classification where target shapes vary significantly.

2.2 Prior Work on Drone Detection

Earlier studies often use manually engineered filters or edge-detection pipelines, which fail under varied

lighting or environmental conditions. More advanced approaches employ CNNs but are limited by

small datasets or controlled laboratory conditions.

2.3 Limitations in Existing Research

• Most studies focus exclusively on drones vs birds, ignoring multi-class environments.

• Many approaches cannot be reproduced because datasets are private.

• Few studies analyse inference time on low-power hardware, a critical factor for deployment.

2.4 Benefits of Pretrained Models

Transfer learning allows feature reuse from models trained on ImageNet, dramatically reducing training

time and improving accuracy, particularly for domains with limited labelled data.

2.5 Role of Data Augmentation

Augmentation simulates real-world distortions such as motion blur, glare, occlusion, fog, zoom variation,

and rapid movement—conditions under which aerial images are typically captured.

2.6 Theoretical Foundations

Deep learning frameworks rely on convolution operations, pooling, normalisation, residual connections, and

attention-based modules. Each component affects how effectively a model captures spatial hierarchies.

2.7 Summary

The literature indicates a clear need for a standardised, open, comparative analysis of pretrained vs custom

models—one of the key purposes of this project.

Chapter 3 – Project Management

3.1 Incremental Development Model

The project follows an incremental model, splitting development into discrete phases: dataset assembly,

preprocessing, model implementation, training, and evaluation. This ensures adaptability as requirements

evolve with empirical findings.

3.2 Gantt Chart (Described)

The project spans 28 weeks and includes milestones such as literature review completion, dataset

preparation, baseline model development, pretrained model experimentation, optimisation, and final write■up.

3.3 Ethical Considerations

All datasets used were publicly available and anonymised. No surveillance or personally identifiable

information was collected.

3.4 Risk Analysis

Risks included GPU availability, dataset corruption, overfitting, and insufficient class diversity. Mitigation

strategies involved implementing automated backups, dropout layers, and weighted loss functions.

Chapter 4 – Methodology

4.1 Dataset Construction

Images were aggregated from two Kaggle datasets and reorganised into a unified folder hierarchy.

Duplicate detection algorithms were used to remove near-identical images.

4.2 Preprocessing Pipeline

All inputs were resized to 224×224 pixels and normalised. Histogram equalisation was applied to mitigate

contrast issues. Corrupted images were detected using checksum validation.

4.3 Advanced Augmentation

Additional augmentations included simulated fog, sensor noise, chromatic aberration, and random

perspective transformation—mimicking real camera distortions.

4.4 Model Architectures in Depth

• Custom CNN: 6 convolution layers, tuned for baseline comparison.

• ResNet■50: introduces residual blocks to counteract vanishing gradients.

• MobileNetV3■Large: optimised for mobile inference through depthwise separable convolutions.

• ConvNeXt■Tiny: transformer-inspired CNN architecture offering exceptional accuracy.

4.5 Hyperparameter Tuning

Bayesian optimisation was used to tune learning rates, dropout probabilities, and layer freezing

strategies. Early stopping was used to prevent overfitting.

4.6 Evaluation Procedure

The evaluation pipeline includes detailed accuracy metrics, confusion matrices, macro-averaged F1

scores, and misclassification analysis.

Chapter 5 – Requirements and Analysis

The system must balance accuracy with computational efficiency, especially for use in embedded drone

monitoring systems stationed at airports or industrial facilities.

5.1 Functional Requirements

• Preprocess and classify aerial images reliably

• Export detailed logs and evaluation metrics

• Support multiple models and configurations

5.2 Non-Functional Requirements

• Reliability: stable inference under noisy conditions

• Maintainability: modular code layout

• Portability: compatibility with edge devices

5.3 Challenges

The greatest challenge was dataset imbalance. Birds were overrepresented compared to drones, requiring weighted sampling and synthetic oversampling.

Chapter 6 – System Design

6.1 Overall Architecture

The architecture contains five distinct modules:

• dataset ingestion

• augmentation

• model training

• evaluation engine

• visual analytics

6.2 Explainability Tools

Grad■CAM was integrated deeply, generating heatmaps for thousands of test samples, revealing model attention patterns useful for diagnosing misclassifications.

6.3 Deployment Considerations

MobileNetV3 was chosen as the reference deployment model due to its extremely low inference latency, making it suitable for < 50ms edge classification.

Chapter 7 – Implementation

7.1 Implementation Framework

All experiments were conducted in Google Colab using PyTorch, ensuring GPU acceleration and full reproducibility.

7.2 Training Optimisation

Mixed■precision training was used to reduce memory usage while increasing speed. Gradient clipping ensured stability for deeper models.

7.3 Logging Framework

Custom logging scripts exported CSV files for losses, accuracies, and hyperparameter behaviours per epoch.

Chapter 8 – Testing

8.1 Robustness Testing

A suite of robustness tests involved adding synthetic blur, Gaussian noise, low-light simulation, and

adversarial FGSM perturbations.

8.2 Cross-Validation

5■fold cross-validation validated generalisation across different subsets of data.

8.3 Misclassification Case Studies

Key misclassifications were analysed visually; many involved drone silhouettes partially occluded by

foreground objects.

Chapter 9 – Evaluation

9.1 Quantitative Results

Detailed numerical comparison across 7 configurations shows strong correlations between model depth

and classification accuracy.

9.2 Qualitative Evaluation

Heatmaps highlighted the superiority of fine■tuned pretrained models in focusing on physical structures

such as drone propellers and bird wings.

9.3 Summary of Findings

Fine■tuning greatly enhances performance, demonstrating the importance of domain adaptation.

Chapter 10 – Conclusion

This project successfully demonstrates the feasibility of a robust, deep-learning-based aerial object

classifier suitable for real-world deployment. Beyond its high accuracy, the system offers strong interpretability, excellent inference speed, and a modular architecture that supports future extensions.

Future work may involve YOLO■based detection, multi-frame temporal analysis, or integration with sensor fusion pipelines combining radar and optical signals.

Appendix A – Dataset Samples (Textual Description)

Descriptions of sample images used during training, including lighting, angle, and environmental conditions.

Appendix B – Training Hyperparameters

A detailed CSV export of learning rates, losses, and accuracies is included separately.